

Climate Data Quality Control Course

2. Course on the Metcap business rules

2.2 FAULTY SENSORS REPORTING PROCEDURES

2.2.1 The quality control procedures of **Metcap** cannot always detect a faulty sensor or does not detect it immediately. The Met Technician doing the quality control of the data must be particularly alert for faulty sensors.

Faulty sensors can usually be identified by absolute wrong values occurring over a number of days.

For example:

- A wind direction greater than 360 degrees
- A temperature value of says 69 °C, and in which case the reading continues to be wrong.

It is the duty of the Met Technician doing the quality control to go back day by day to determine when the sensor became faulty. He/she must do so, irrespective whether or not such readings come under their area of responsibility. All faulty data must be deleted and the day's quality controlled so that the data in the hourly and daily tables can be corrected.

Once it has been determined when the sensor became faulty, the following actions must be taken:

• From the time that the sensor became faulty up to and until it is repaired, replaced or disabled	All data must be deleted.
• The sensor must be disabled until such time that it can be repaired or replaced.	Note: if you deactivate a sensor, data that is already on the database from that sensor will not be deleted by this action. If all sensors at the station need to be deactivated please remember to quality control the missing data.
• Upon becoming aware of a non-operational AWS/ARS or a sensor/s at an AWS.	A Regional Office must inform Climate Services as soon as possible.
• Notwithstanding the above, a non-operational station must be visited within three working days after the discovery of its nonoperational sensor. Exceptions are where stations are not owned by SAWS or are in locations that are difficult to access.	Thus the Regional Office must be equipped with an adequate number of spares to ensure proper maintenance of all its AWS and ARSs.

2.3 FAULTY SENSORS METCAP TESTING AND QC PROCEDURES

2.3.1 Faulty sensors: Identical readings

- Consecutive identical readings may indicate a faulty instrument or sensor, but it is not necessary the case. - Under certain weather conditions consecutive identical readings are possible. - The Met Technician must use his/her knowledge and skills to determine whether identical readings are caused by faulty instruments or weather phenomena. - If the Met Technician is uncertain, he/she must consult his supervisor or other senior staff members. - The following tests on consecutive readings are done and carried out by Metcap. - Also see the test for wind paragraph 2.3.7.	
2.3.1.1 Humidity	- Eighty (80) consecutive readings are tested. - These readings are identified if they are identical. Met Technician to verify.
2.3.1.2 Pressure	- Forty (40) consecutive readings are checked. - These readings are identified if they are identical. Met Technician to verify.
2.3.1.3 Temperature	- Twenty (20) consecutive readings are tested. - These readings are identified if they are identical. Met Technician to verify.

2.3.2 Faulty sensors: Sharp drop or rise

- A sharp drop or rise can also indicate a faulty sensor especially if this keeps occurring at different time intervals over a period of time. - Two consecutive readings are compared. - When the difference between the two readings exceeds a predefined limit it is indicated as a sharp drop (decrease) or rise (increase). It is up to the Met Technician to determine if this is a meteorological or non-meteorological occurrence and act accordingly. - The following tests are done by Metcap.	
2.3.2.1 Humidity	If the difference between two consecutive readings exceeds 20%, it is indicated as a sharp drop or rise as the case may be.
2.3.2.2 Pressure	Difference between two consecutive reading must not exceed 2.0 hPa
2.3.2.3 Temperature	The difference between two consecutive readings must not exceed 5.0°C.

2.3.3 Faulty sensors: Small range tests

- This is almost similar to consecutive identical readings with the exception that small changes in the values might occur. - A small data range over a period of time may indicate a faulty instrument, but it is not necessary the case. - The Met Technician must use his/her knowledge in making a decision. (See 2.3.1 identical readings). - Tests for small ranges are carried out over the whole of a day, i.e. a 24 hour period or 288 consecutive 5 minute readings. - The predefined limits for each weather elements are as follows:	
2.3.3.1 Humidity	- The lowest and highest reading between 00:05SAST and 00:00SAST (24 hour period) must be greater than or equal to 5%. Note: although the time readings from the AWS logger are Universal Time (Greenwich Mean Time), Metcap converts to SAST before testing the data.
2.3.3.2 Pressure	The lowest and highest reading between 00:05SAST and 00:00SAST (24 hour period) must be greater than or equal to 1.0 hPa.
2.3.3.3 Temperature	The lowest and highest reading between 00:05SAST and 00:00SAST (24 hour period) must be greater than or equal to 0.5°C.

2.3.4 Faulty sensors: Spikes

- Spikes occur when there is a sharp rise or drop from two consecutive readings and the third consecutive reading is the same or nearly the same as the first.
- Spikes can occur regularly in the data.
- It is mostly caused by external factors such as lightning strikes close to the AWS/ARS station.
- However, if spikes occur frequently it can be an indication of a problematic sensor, or a problem with the battery voltage.

2.3.4.1 Humidity	• Three consecutive readings are tested namely x, y and z. • A spike is identified if the difference between x and y is greater than 10% and the difference between y and z is less than 5%.
2.3.4.2 Pressure	• Three consecutive readings are tested namely x, y and z. • A spike is identified if the difference between x and y is greater than 1.0 hPa and the difference between y and z is less than 0.5 hPa.
2.3.4.3 Temperature	• Three consecutive readings are tested namely x, y and z. • A spike is identified if the difference between x and y is greater than 2.0°C and the difference between y and z is less than 1.0° C.

2.3.5 Faulty sensors: Limits

- These are checks to ensure that the climate elements being tested are within acceptable predetermined limits.
- In these tests the limits have been subdivided into certain degrees of errors.
- What this means is that if a measured climatological parameter is outside its normal range but within an acceptable amount outside the range it is automatically corrected by Metcap.
- If it falls totally outside the range the value is deleted.
- The reason for this is to allow for the accuracy specifications of the sensors, i.e. a humidity sensor can measure humidity with accuracy of +/- 5%.
- As stated before where Metcap automatically changes values a corresponding message is given.

2.3.5.1 Humidity Tests	• Values greater than 100% but less than 105% are automatically changed to 100%. • Values greater than 105% or less than 1% are automatically deleted.
2.3.5.2 Pressure Tests	• Values less than 800.0 hPa or greater than 1050.0 hPa are automatically deleted.
2.3.5.3 Temperature Tests	• Values less than -20.0°C or greater than 45.0°C are checked and a warning message is produced. • <i>Please note that these values are not automatically deleted as they may in fact be valid</i>

Tip: As a Met Tech do you know the temperature and pressure ranges of the stations under your watch?
 A good idea to create a table with the maximum and minimum values per station for easy reference.
 Climate Data Unit can provide you with this information.

2.3.6: Missing readings tests

- As with spikes, missing readings can indicate a faulty sensor if they occurs frequently for the same sensor.
- Similarly, if it occurs frequently from the same AWS the error may be in the communications with the station.
- These situations must be monitored and all efforts must be made to rectify the situation. This can be done by downloading data from the logger again.
- Except for wind measurements and rainfall measurement, the action taken by Metcap is the same for all parameters. Refer to paragraph 2.1.1.

2.3.6.1 for all Temperature, Humidity and Pressure readings

- A test is done to ensure all 5 minute data between 00:05 SAST and 00:00 SAST (24 hour period) are downloaded.
- A re-download is executed if not successful.
- Metcap tests for missing readings on the hour and where at least one of the two readings before the hour and at least one of the two readings after the hour are not missing the missing hourly data will be interpolated. (See Paragraph 2.1.1)

2.3.7: Rainfall Tests

- Rainfall reported in the 06:00Z SYNOP must be checked by the Met Technician.
- The quality control component of Metcap cannot check the validity of the total rainfall, except in the given circumstances below.

- Rainfall reported in the 06:00Z SYNOP must be checked against the rainfall from the standard rain gauge or AWS.
- Other rainfall values from remote AWS station must be checked against rainfall obtained from surrounding stations, manual as well as AWS or ARS.
- Only two tests are done by Metcap on rainfall data.
 - Values less than 0.0 mm is automatically deleted.
 - If values greater than 10.0mm in a 5 minute period are measured Metcap warns the Met Technician, but **DOES NOT** automatically deletes the data. It is up to the Met Technician to verify as this value could be correct.

2.3.8: Wind Direction; Speed and Gust

2.3.8.1 Wind Direction.

- When three or more 5 minute values between any two hours are missing the wind direction for the hour is not calculated.
- The reason for this is that the hourly wind is the mean wind direction for the full hour.
- If the wind direction reading is deleted, the corresponding vector speed and vector deviation readings are also deleted.
- The corresponding hourly wind speed and wind gust is not deleted if they are not faulty.
- Wind Direction limits test:
 - Values less than 0 degrees and more than 360 are deleted.
- Wind Direction small range test:
 - Wind direction readings between 00:05 SAST and 00:00 SAST (24 hour period) must be greater or equal than 5 degrees.
- Wind direction identical tests:
 - Forty identical readings are tested.
 - These readings are identified if they are identical. Met Technician to verify.

2.3.8.2 Wind Gust

- Wind Gust: Limits tests:
 - Values less than 0.0 ms⁻¹ or greater than 120 m.s⁻¹ are automatically deleted.
- Wind Gust: Small range tests:
 - The lowest and highest reading between 00:05 SAST and 00:00 SAST (24 hour period) must be greater or equal than 3.0 m.s⁻¹.
- Wind Gust: Strong Gust tests:

	- Values greater than 32.7 m/s m.s⁻¹ and less than 120.0 m/s m.s⁻¹ are tested and a warning message is produced. - The values are not automatically deleted. <u>Wind Gust and Average Wind Speed Comparison Tests:</u> - Test 1 (Speed less or equal than 20.0m/s m.s⁻¹). - Where wind speed average is less than 20.0 m/s m.s⁻¹ the difference between the average winds speed and the wind gust must not be greater than 20.0 m/s m.s⁻¹. <u>Wind Gust and Average Wind Speed Comparison Tests:</u> - Test 2 (Speed greater than 20.0 m/s m.s⁻¹). Where wind speed average is greater than 20.0 m/s m.s⁻¹ the difference between average wind speed and the wind gust must not be greater than 25.0 m/s m.s⁻¹.
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2.3.9: Battery Voltage Tests	
- Although Battery Voltage is not a climatological parameter, it is of critical importance. - A too low or to high battery voltage will/could result in wrong recordings of climate data by the climatological sensors in the AWS/ARS. - It is therefore necessary to test this value as well. - It is a fact than a lot of the errors described above can be because of incorrect battery voltage.	
2.3.9.1 Battery Voltage: <u>Limits Test 1:</u>	- Test for battery voltage greater than 13.6 volts dc. - This usually indicated there is something wrong with the AWS/ARS. - The Met Technician must under these circumstances pay extra attention to the climatological values received from the AWS/ARS as it may be erroneous.
2.3.9.2 Battery Voltage: <u>Limits Test 2:</u>	- Test for battery voltage greater than or equal to 11.0 volts dc and less than 12.5 volts dc. - This is a warning and indicates that something is busy going wrong with the AWS. - Monitor this AWS carefully; if necessary visit the AWS in order to prevent possible data loss. - Check climate data carefully as it may be faulty.
2.3.9.3 Battery Voltage: <u>Limits Test 3:</u>	- Test for battery voltage greater than or equal to 9.6 volts dc and less than 11.0 volts dc. - All pressure values are automatically deleted as it is no longer reliable. - Other climate parameters may still be correct. - This must be carefully checked.
2.3.9.4 Battery Voltage: <u>Limits Test 4:</u>	- Test for battery voltage less than 9.6 volts. - All data are deleted, no longer reliable.

2.4 Quality control of data via Metcap

Met Technicians using Metcap must be aware that this software is designed to interact with a number of tables on ClimDB. This means that all steps in the process of quality controlling data must be performed each and every time to make sure that the data is quality controlled on all the tables. The same principle applies when dumping data. If this is not done correctly data might not be sent to all the tables on ClimDB. Detailed work instructions for capturing, quality controlling and sending data via Metcap are available on the EQMS under climate documents (doc ref: CLS-WI-QC-Metcap.2).